Emma Kadlecek 08/21/2026 Per 6

1. $5x + 2y = 11$ and $x + y = 4$
   $5x + 2(x + 4) = 11$
   $5x + 2x + 4 = 11$
   $7x + 4 = 11$
   $7x = 7$
   $x = 1$
   $1 + y = 4$
   $y = 3$
2. $6x - 7y = 47$ and $2x + 5y = -21$
   $6x - 7y = 47$ and $6x + 15y = -63$
   $-22y = 110$
   $y = -5$
   $6x - 7(-5) = 47$
   $6x + 35 = 47$
   $6x = 12$
   $x = 2$
3. $5x + 7y = -15$ and $2x + 9y = -6$
   $10x + 14y = 30$ and $-10x + -45y = -30$
   $31y = 0$
   $y = 0$
   $5x + 7(0) = -15$
   $5x + 0 = -15$
   $5x = -15$
   $x = -3$
4. $2x + 3y = 2$ and $4x - 9y = -1$
   $8x + 12y = 8$ and $8x - 18y = -2$
   $-6y = 6$
   $y = -1$
   $2x + 3(-1) = 2$
   $2x + -3 = 2$
   $2x = 5$
   $x = 2.5$
5. $9x - 7y = 5$ and $10x + 3y = -16$
   $-27x + 21y = -15$ and $70x + 21y = -112$
   $-97x = 97$
   $x = -1$
   $9(-1) - 7y = 5$

$-9-7y=5$

$-7y=14$

$y=-2$

6. $14x+31y=-6$ and $21x+17=50$

   $42x+93y=-18$ and $-42+-34=-100$

   $59y=-118$

   $y=-2$

   $14x+31(-2)=-6$

   $14x+-62=-6$

   $14x=56$

   $x=4$

7. $13x+10y=-7$ and $17x-15y=47$

   $39x+30y=-21$ and $34x-30y=94$

   $73x=73$

   $x=1$

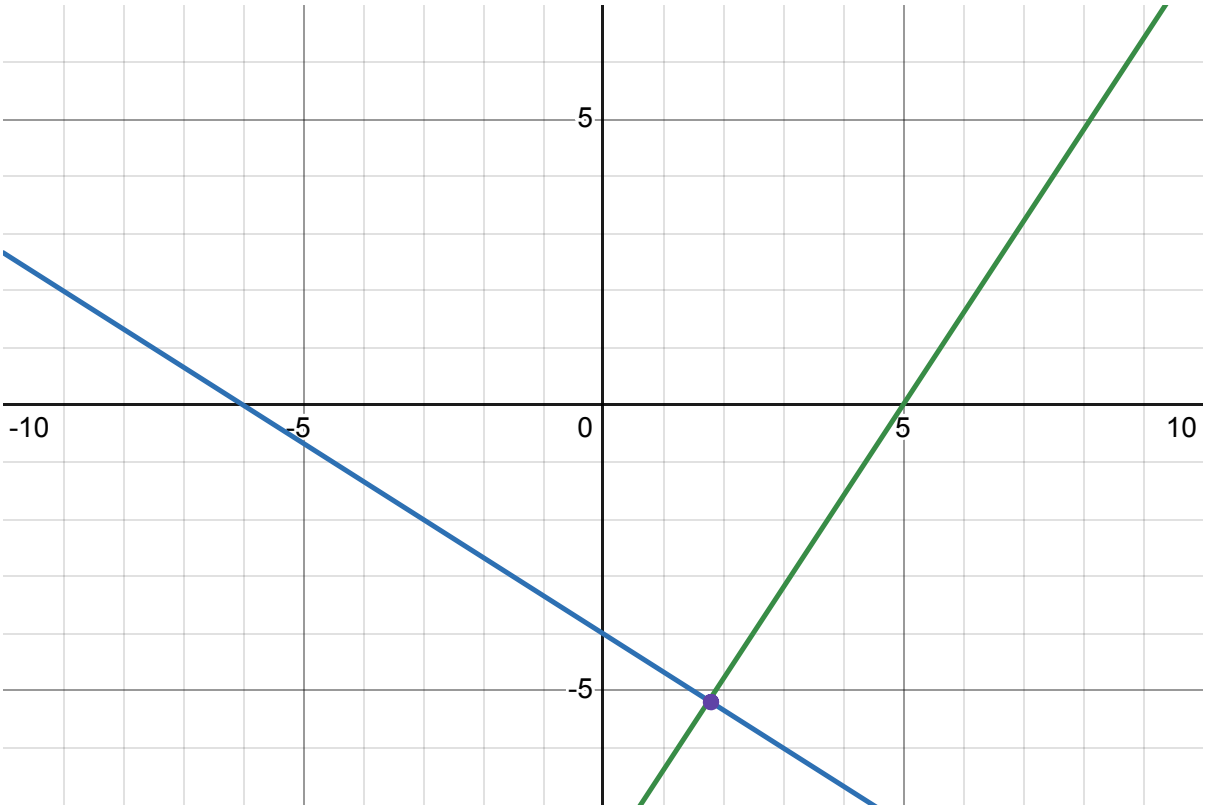
   $13(1)+10y=-7$

   $13+10y=-7$

   $10y=-20$

   $y=-2$

8. ```
   $(1.8,-5.2)$
   $2(1.8)+3(-5.2)=-12$
   $3.6 + -15.6=-12$
   $8(1.8) - 5(-5.2)=40$
   $14.4--26=40$
   ```
   ```



9. $x + 2y = 2$ and $5x - 3y = -29$

   $5(2y - 2) - 3y = -29$
   $10y - 10 - 3y = -29$
   $13y - 10 = -29$
   $13y = -39$
   $y = 3$
   $x = 2 - 2(3)$
   $x = 2 - 6$
   $x = -4$

10. $2x - 9y = 14$ and $6x - y = 42$

    $2x - 9(6x - 42) = 14$
    $2x - 54x + 378 = 14$
    $-52x + 378 = 14$
    $-52x = -364$
    $x = 7$
    $y = 6(7) - 42$ HAHAHAHAHAHAHAHAHAHA 67
    $y = 42 - 42$
    $y = 0$

11. $x - 3y = 13$ and $5x + 3y = 2$

    $5(13 + 3y) + 3y = 2$
    $65 + 18y = 2$
    $18y = -63$
    $y = -3.5$
    $x - 3(-3.5) = 13$
    $x + 10.5 = 13$
    $x = 2.5$